Explicit Pressure and Local Contraction Bounds for a Forced Navier–Stokes Blow-Up Construction

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Abstract

We collect explicit sufficient bounds for two local components of the September 2026 forced Navier–Stokes construction attributed to OpenAI. An integrable envelope for its unbumped outer schedule gives pressure majorants $16\overline{P}^2$ and $128\overline{P}^2$ on a fixed complex rectangle. For the leading axis equations, conservative coefficient estimates give a finite, noncircular rule selecting the contraction parameter and then the amplitude. The resulting fixed-point map has displacement and Lipschitz constant at most $1/16$ on a unit ball, and its real normalized angular profile is at least $1/6$ on the stated interval. These are conditional quantitative refinements of the source's existing local argument. They do not select globally admissible outer data, certify the completed velocity field, or provide a computational complexity result.

1 Contribution, provenance, and scope

The external manuscript [1] already introduces the pressure datum, weighted analytic space, radial inverse estimates, and local contraction. Our purpose is to expose sufficient constants and their order of dependence. In particular, this note does not claim a new blow-up theorem or a new fixed-point method. No priority claim is made for the general techniques. The source-specific estimates are candidates for a quantitative follow-up; a focused literature screen is not a certification of novelty.

The derivations were developed in the research records listed in the repository's provenance manifest. This manuscript consolidates the pressure, coefficient, and contraction records from one fixed commit. Its mathematical arguments are informal. Existing AI-assisted reviews concern those records, not independent human peer review or a machine-checked proof of this note. The source manuscript is used as an external mathematical input, not independently certified here. Appendix labels below refer to its 166-page version retrieved on September 8, 2026; its checksum is recorded separately.

2 A pressure envelope on a fixed complex domain

Let $P_* > 0$, and let $c : \mathbb{R} \to (0, \infty)$ and $\theta : \mathbb{R} \to [0, 1]$ be measurable. Assume the envelope

$$c(y) \le P_* \begin{cases} e^{y/10}, & y \le 1, \\ e^{1/10} e^{-(y-1)/2}, & y \ge 1. \end{cases} \tag{1}$$

These assumptions isolate exactly the radial estimate needed here. For the source's unbumped schedule, $c'/c = l - 1/2$: the inner segment has $l = 3/5$, the first unit interval has $l \le 3/5$, and subsequent segments have $l \le 0$ at the central angular parameter. Integrating those inequalities gives (1). This statement concerns the reference schedule in the datum (A.21), not arbitrary later edited profiles or the complete pressure.

Define

$$\Pi_0(z) = -\frac{1}{2}\int_{\mathbb{R}} c(y)^2 \exp(-2\theta(y)\,\mathrm{Log}(1+z^2))\,dy, \quad \Omega = \{z : |\,\mathrm{Re}\, z| < 5/4,\ |\,\mathrm{Im}\, z| < 1/4\}. \tag{2}$$

The logarithm is the principal branch.

Theorem 1. *Under* (1), Π_0 *is holomorphic on* Ω. *For every rational* $\overline{P} \ge P_*$,

$$|\Pi_0(z)| \le 16\overline{P}^2 =: P_0, \qquad |\Pi_0'(z)| \le 128\overline{P}^2 =: P_1 \quad (z \in \Omega).$$

Proof. The three intervals $(-\infty, 0)$, $[0,1]$, and $[1,\infty)$ contribute at most $5P_*^2$, $e^{1/5}P_*^2$, and $e^{1/5}P_*^2$, respectively, to $\int c^2$. Since $e^{1/5} \le \sum_{k\ge 0}(1/5)^k = 5/4$,

$$\int_{\mathbb{R}} c^2 \le (5 + 2e^{1/5})P_*^2 < 8P_*^2.$$

On Ω, $\mathrm{Re}(1+z^2) > 15/16$ and $|z| < 3/2$. Consequently $f = (1+z^2)^{-1}$ satisfies $|f| < 2$ and $|f'/f| < 4$. For real $0 \le \theta \le 1$, the integrand's power has magnitude at most 4 and its z derivative has magnitude at most 32. These bounds give common integrable majorants. Holomorphic integration and differentiation under the integral are therefore valid, and multiplication by the factor $1/2$ proves the two bounds. □

For $R \ge 1$, the pressure contribution outside $[-R, R]$ is at most

$$10\overline{P}^2 e^{-R/5} + 3\overline{P}^2 e^{-(R-1)}. \tag{3}$$

The corresponding derivative tail is at most eight times this expression. This follows by integrating the two tails in (1) and using the same power bounds. For a finite schedule with computable parameters and effective compact-interval moduli, certified quadrature and (3) give finite-accuracy pressure evaluation. Measurability and the envelope alone do not imply computability. No quadrature implementation or cost bound is asserted here.

3 Coefficient domain and explicit local data

Fix rational parameters

$$0 < h \le 1/100, \quad 0 < j \le 1/20, \quad 0 < \sigma \le 1, \quad 0 \le e \le 1/1000,$$

and choose rational $0 < r \le \min(1/1000, \sigma^2/4096)$. Write $I = [-1-e, 1+e]$, let Ω_r be its open complex r-neighborhood, and put $\rho = r/4$. Define

$$\begin{aligned} &A = 1/2 + h, && D = 1/2 - h, && d = 1 - \eta^2, && L = 1 - 2h\eta^2, \\ &U_* = 4\eta + j, && H_* = D\eta + dU_*, && Q_* = H_*^2 + \sigma^2, && \\ &\chi = H_*^2/Q_*, && \zeta = -LH_*/Q_*. && && \end{aligned}$$

$$W_* = 1 - dU_*' - 2D\eta U_*, \qquad Z_* = -A(1-2\eta U_*)U_* - H_*U_*' - d\Pi_0' + 4A\eta\Pi_0.$$

These are the coefficients in the source's Appendix B. The source's global outer admissibility and additional separation condition (B.2) are not established by the parameter ranges above. Those remain additional obligations when applying the local construction to the complete source.

Lemma 2. *On Ω_r one has*

$$|H_*| < 9, \quad |H_*'| < 18, \quad 1/2 < |L| < 2, \quad |Q_*| \geq \sigma^2/2.$$

Furthermore,

$$|\chi| \leq 162/\sigma^2, \quad |\zeta| \leq 36/\sigma^2, \quad |Z_*/L| \leq \overline{Z} := 192 + 928\overline{P}^2.$$

Proof. The tube lies in $|z| < 101/100$. Triangle bounds for the displayed polynomials give the first three inequalities and $|(H_*^2)'| < 324$. For each z in the tube there is real $x \in I$ with $|z - x| < r$. Since $Q_*(x) \geq \sigma^2$, integration along the segment gives $|Q_*(z)| \geq \sigma^2 - 324r > \sigma^2/2$. This proves the bounds for χ and ζ. For Z_*, use $|U_*| < 5$, $|d| < 3$, $|1 - 2zU_*| < 12$, $|H_*U_*'| < 36$, $A < 1$, and $|4Az| < 5$. It follows that $|Z_*/L| \leq 192 + 10P_0 + 6P_1 = \overline{Z}$. The entire tube lies inside Ω, so Theorem 1 applies. □

For $\Lambda \geq 1$ set

$$\phi_*(z) = \exp\left(\Lambda \int_0^z \zeta(w)\, dw\right), \qquad g = \phi_*/C.$$

The tube is convex, the path length is less than two, and hence $|\phi_*| \leq e^{72\Lambda/\sigma^2}$ there. In particular,

$$C \geq 2^{\lceil 144\Lambda/\sigma^2 \rceil} \quad \Longrightarrow \quad |g| \leq 1 \text{ on } \Omega_r, \tag{4}$$

because $\log 2 > 1/2$. A bound on this complex domain, rather than only on its real interval, is needed to control coefficient derivatives.

4 The weighted space and operator constants

We use the source's space with the following normalization. Write $F(Y, \eta) = \sum_{\alpha \geq 0} F_\alpha(\eta) Y^\alpha$ and put

$$a_{\alpha\beta} = \frac{20^{-\alpha}\rho^{-\beta}\beta!\binom{\alpha+\beta}{\beta}}{(\alpha+1)^2(\beta+1)^2}, \qquad \|F\|_\rho = \sup_{\alpha,\beta \geq 0,\ \eta \in I} \frac{|\partial_\eta^\beta F_\alpha(\eta)|}{a_{\alpha\beta}}.$$

The space B_ρ consists of smooth coefficient sequences with finite norm and is complete, by uniform convergence at every coefficient and derivative order. On B_ρ^2 use the maximum norm. Set

$$\mathcal{I}F = \int_0^Y F(s,\eta)\, ds, \quad \mathcal{A}F = Y^{-1}\mathcal{I}F, \quad \mathcal{D} = Y\partial_Y, \quad (\mathcal{J}_\nu F)_{\alpha+1} = \frac{F_\alpha}{(\alpha+1)(\alpha+\nu)}, \quad (\mathcal{J}_\nu F)_0 = 0.$$

Here $\nu = 1, 2$; averaging is defined at zero by its coefficient extension.

The coefficient convolution and weight shifts in Lemma B.1 give

$$\begin{aligned} a_{\text{alg}} = 256, \quad \|\mathcal{J}_\nu\|, \|\mathcal{I}\|, \|Y \cdot\| \leq 80, \quad \|\partial_\eta \mathcal{I}\| \leq 80/\rho, \quad \|\mathcal{A}\| \leq 1, \\ \|\mathcal{J}_\nu[(\partial_\eta F)(\mathcal{D}G)]\|_\rho \leq C_{\text{mix}} \|F\|_\rho \|G\|_\rho, \qquad C_{\text{mix}} = 20480/\rho. \end{aligned} \tag{5}$$

For completeness, the convolution constant in each index is bounded by $8\sum_{i \geq 1} i^{-2} < 16$, giving 256 for multiplication. The weight ratios satisfy

$$\frac{a_{\alpha\beta}}{a_{\alpha+1,\beta}} \leq 80, \qquad \frac{a_{i,k+1}}{a_{i+1,k}} \leq \frac{80}{\rho}(i+1).$$

For a mixed term at output radial degree $N = \alpha + 1$, assign the raised degree to the factor carrying ∂_η. The residual factor after radial inversion is $(i+1)(\alpha - i)/[N(\alpha + \nu)] \leq 1$. If the radial derivative is omitted, replace $\alpha - i$ by one. Without the parameter derivative, the first weight ratio gives $80(\alpha - i)/[N(\alpha+\nu)] \le 80$; omitting both is easier. Thus the same C_{mix} covers all these cases, including $\alpha = 0$. Averaging costs no factor. Extra undifferentiated coefficients are handled by further convolution, each at cost 256; they are not pulled through radial inversion. This does not assert boundedness of either unintegrated derivative alone.

For an η-only function bounded by M on Ω_r, Cauchy estimates on disks of radius $r/2$ give

$$\|a\|_\rho \le M \sup_{\beta \ge 0} \frac{(\beta+1)^2}{2^\beta} \le 3M.$$

Consequently multiplication by χ has norm at most $M_\chi = 124416/\sigma^2$. If $T_\chi = \mathcal{J}_2\chi/2$, where multiplication precedes integration, its radial degree increases at every application. The sharper bound on inputs with no degree below b is $\|\mathcal{J}_2 F\| \le 80\|F\|/[(b+1)(b+2)]$, so

$$\|T_\chi^k\| \le \frac{(40M_\chi)^k}{k!(k+1)!}, \qquad \|(1+T_\chi)^{-1}\| \le e^{40M_\chi} \le V := 2^{\lceil 80M_\chi \rceil}.$$

The inverse follows from the absolutely convergent alternating operator series. It does not require $\|T_\chi\| < 1$.

5 Integrated remainders and a contraction rule

Set

$$\begin{gathered} K = \max(V, 120\overline{Z}) + 1, \qquad B_c = 1000/\sigma^2, \\ T = a_{\mathrm{alg}} B_c C_{\mathrm{mix}}, \qquad P_c = 57600 a_{\mathrm{alg}}^4 B_c (2 + 1/\rho), \\ N = T(6K + 11K^2) + P_c K^2, \qquad L_R = T(6 + 22K) + 2P_c K. \end{gathered} \tag{6}$$

All these constants are fixed before Λ and C. The center is

$$\Phi_0 = f_0(Y\chi), \quad u_0 = -YZ_*/(2L), \qquad f_0(z) = \sum_{\alpha \ge 0} \frac{(-z/2)^\alpha}{\alpha!(\alpha+1)!}.$$

Its component norms are at most V and $120\overline{Z}$, respectively, so the closed unit ball about it has both component norms at most K.

Write $\varepsilon = \Lambda^{-1}$ and define

$$B = -2D\eta\mathcal{A}u - d\partial_\eta \mathcal{A}u, \quad W = W_* + \varepsilon B, \quad U = U_* + \varepsilon u, \quad H_c = H_* + \varepsilon du, \quad p = \mathcal{I}(g^2\Phi^2).$$

The source remainders, with their full derivative dependence, are

$$LR_1 = [W + h(1 - 2\eta U) + du\zeta]\Phi + W\mathcal{D}\Phi + H_c\partial_\eta\Phi, \tag{7}$$

$$\begin{aligned} LR_2 = {} & [A(1 - 4\eta U_*) + dU_*']u - 2A\eta\varepsilon u^2 + W\mathcal{D}u + H_*\partial_\eta u + \varepsilon du\partial_\eta u \\ & - 4A\eta p + d\partial_\eta p - 2\eta\mathcal{D}p. \end{aligned} \tag{8}$$

Expanding these expressions gives three linear and seven quadratic nonpressure terms in R_1, and three linear, four quadratic, and three pressure terms in R_2. Appendix A lists them individually.

Every η-only coefficient in that list, including its division by L, has norm at most B_c. For example $|W_*| < 19$ on the tube, and $\|d\zeta/L\|_\rho \le 648/\sigma^2$. The axial linear coefficient before division by L has magnitude below 34; Cauchy conversion bounds its norm after division by $204 < 210$. The other coefficients obey smaller bounds by Lemma 2. After applying the indicated $\mathcal{J}_\nu$, (5) therefore gives TK per nonpressure linear term and TK^2 per quadratic term. The powers of ε are at most one.

By (4), $\|g\|_\rho \le 3$. For one fixed g per contraction, four-factor multiplication gives

$$\left\|g^2\Phi^2\right\|_\rho \le 9a_{\mathrm{alg}}^3K^2, \qquad \left\|g^2(\Phi^2-\Psi^2)\right\|_\rho \le 18a_{\mathrm{alg}}^3K\,\|\Phi-\Psi\|_\rho\,.$$

The operators generating p, $\partial_\eta p$, and $\mathcal{D}p$ from $g^2\Phi^2$ are $\mathcal{I}$, $\partial_\eta\mathcal{I}$, and multiplication by Y. Applying these first, then coefficient multiplication and the outer $\mathcal{J}_\nu$, gives the total pressure bound $80\cdot 80\cdot 9a_{\mathrm{alg}}^4B_c(2+1/\rho)K^2 = P_cK^2$. Replacing the two unknown factors one at a time gives its difference bound $2P_cK$ times the input difference. Summing both component bounds proves that N and L_R in (6) bound the norm and Lipschitz constant of $(\mathcal{J}_2R_1, \mathcal{J}_1R_2)$ on the unit ball.

Theorem 3. *For the local data and pressure above, choose an integer*

$$\Lambda \ge \max\{1, \lceil 8V(N+L_R+1)\rceil\},$$

and then choose C satisfying (4). The map

$$\mathcal{F}(\Phi,u) = \left(\Phi_0 + \frac{(1+T_\chi)^{-1}\mathcal{J}_2R_1}{2\Lambda}, \quad u_0 + \frac{\mathcal{J}_1R_2}{2\Lambda}\right)$$

maps the closed unit ball about (Φ_0,u_0) into itself and has Lipschitz constant at most $1/16$. Its unique fixed point in that ball has distance at most $1/16$ from the center. On $0\le Y\le 41/10$, $\eta\in I$, its real angular component satisfies $\Phi \ge 1/6$.

Proof. The map's displacement and Lipschitz constant are bounded by $VN/(2\Lambda)$ and $VL_R/(2\Lambda)$, respectively, each at most $1/16$. Banach's theorem applies on the complete closed ball. Although g depends on the later choices, its common norm bound follows from the same larger complex domain for every allowed pair (Λ, C). Thus selecting Λ using these constants is noncircular.

The radial-zero conditions $\Phi(0,\eta)=1$ and $u(0,\eta)=0$ are preserved: both radial inverses have zero constant coefficient, and the angular inverse preserves this subspace term by term. Real data and iteration from the real center give real coefficients. For $\beta=0$, the weights imply $|F(Y,\eta)| \le \|F\|_\rho/(1-|Y|/20)$, whose factor is below $4/3$ on the stated interval. Also $0\le\chi\le 1$ on I. The alternating series for f_0 gives, with $t=z/2$,

$$f_0(z) \ge 1 - t/2 + t^2/12 - t^3/144 \ge 305719/1152000 > 1/4 \quad (0\le z\le 41/10).$$

The cubic is decreasing on $[0, 41/20]$; the remaining alternating tail starts positively with decreasing magnitudes. Therefore $\Phi \ge 1/4 - (4/3)(1/16) = 1/6$. □

The coefficient binomial sum also gives convergence on $|Y|\le 5$ with complex η displacement less than $\rho/4$ from the real interval. Indeed the geometric factors have sum at most $1/4+1/4<1$ before the polynomial weight denominators are used. In the original radial variable this establishes a leading local profile on $0\le X\le 41/(10\Lambda)$. It does not establish the source's later continuation or endpoint shear inequalities.

6 Effectivity, verification, and remaining obligations

Exact Picard iterates starting at the center have first displacement at most 1/16 and successive increments bounded by powers of 1/16. After m iterations the norm error is at most $(1/16)^{m+1}/(1-1/16)$. This is a bound for exact analytic iterates. Finite coefficient truncations need not converge in the full weighted supremum norm: their tails can continue to attain that norm. On smaller compact analytic domains, the coefficient envelope instead gives effective value and fixed-derivative tails. With effective pressure and parameter data this supports finite-accuracy local evaluation in principle; an implementation must account for arithmetic error, coefficient access, and both truncation and iteration errors.

The constants are deliberately loose. Writing a power of two as an expression is distinct from writing its full binary expansion, and the number of exact Picard steps is distinct from the work needed to evaluate each step. Neither a polynomial bit-time bound nor an implemented certified local solver is provided.

The companion exact-rational script checks elementary numerical inequalities used in these estimates. It is not a formal proof of the analytic argument. The archived reviews separately address source alignment and the conditional claims; the consolidated manuscript has not received independent human peer review.

To obtain a certified completed field one would still need globally admissible outer parameters, the source's separation and continuation conditions, higher-order corrections, and normalized cutoff-tail bounds. Theorems 1 and 3 do not discharge those obligations. They also do not construct logic, memory, a readable fluid device, or a SAT algorithm. The intended contribution is the explicit local certificate and its auditable dependence on the stated inputs.

A Expanded remainder inventory

Each coefficient below includes division by L. A dash in the last column means a multiplier of one. This inventory fixes the term counts used in (6).

Coefficient in R_1	Unknown factors	Multiplier
$(W_* + h(1-2\eta U_*))/L$	Φ	–
$d\zeta/L$	$u\Phi$	–
W_*/L	$\mathcal{D}\Phi$	–
H_*/L	$\partial_\eta\Phi$	–
$-2D\eta/L$	$(\mathcal{A}u)\Phi$	ε
$-d/L$	$(\partial_\eta\mathcal{A}u)\Phi$	ε
$-2h\eta/L$	$u\Phi$	ε
$-2D\eta/L$	$(\mathcal{A}u)\mathcal{D}\Phi$	ε
$-d/L$	$(\partial_\eta\mathcal{A}u)\mathcal{D}\Phi$	ε
d/L	$u\partial_\eta\Phi$	ε

Coefficient in R_2	Unknown expression	Multiplier
$[A(1-4\eta U_*)+dU_*']/L$	u	–
W_*/L	$\mathcal{D}u$	–
H_*/L	$\partial_\eta u$	–
$-2A\eta/L$	u^2	ε
$-2D\eta/L$	$(\mathcal{A}u)\mathcal{D}u$	ε
$-d/L$	$(\partial_\eta \mathcal{A}u)\mathcal{D}u$	ε
d/L	$u\partial_\eta u$	ε
$-4A\eta/L$	p	–
d/L	$\partial_\eta p$	–
$-2\eta/L$	$\mathcal{D}p$	–

Author and assistance disclosure

Daniel Fredriksen directed the research and publication preparation. OpenAI Codex assisted with derivations, internal reviews, consolidation, and repository preparation. AI-assisted checks are not human peer review; author responsibility and source attribution remain explicit.

References

[1] OpenAI. *Finite time blowup for Navier–Stokes*. September 2026 manuscript, 166 pages. Appendix A.2–A.4, especially (A.21), and Appendix B.1–B.3, especially (B.4)–(B.16). Source PDF. Retrieved September 8, 2026. Exact downloaded-file checksum and derivation provenance appear in the companion repository.

[2] Daniel Fredriksen. *Navier–Stokes effective bounds: companion sources and verification*. `https://github.com/Quantyra/navier-stokes-effective-bounds`.